CSV Reading & Writing - Simple Guide

📦 Import (Always)

```
import csv
```

📖 READING

Basic Reading

```
with open("data.csv", "r") as file:
    reader = csv.DictReader(file)  # Creates reader

    for row in reader:              # Loop through each row
        print(row['column_name'])   # Access by column name
```

Save for Later

```
with open("data.csv", "r") as file:
    data = list(csv.DictReader(file))  # Save ALL rows

# Now use anywhere
print(len(data))            # Count rows
print(data[0]['company'])   # First row, 'company' column
```

⚠️ IMPORTANT: Numbers are Strings!

```
# WRONG - can't add strings
total = row['price'] + row['tax']

# RIGHT - convert first
total = float(row['price']) + float(row['tax'])
```

✏️ WRITING

Create New File

```
data = [
    {'name': 'Alice', 'age': '30'},
    {'name': 'Bob', 'age': '25'}
]

with open("people.csv", "w", newline="") as file:
    writer = csv.DictWriter(file, fieldnames=['name', 'age'])
    writer.writeheader()     # Write column names
    writer.writerows(data)   # Write all rows
```

Add to Existing File

```
new_person = {'name': 'Charlie', 'age': '35'}

with open("people.csv", "a", newline="") as file:  # "a" = append
    writer = csv.DictWriter(file, fieldnames=['name', 'age'])
    writer.writerow(new_person)  # NO writeheader() here!
```

📁 File Modes

Mode	What it does	Use when...
`"r"`	Read	Loading data
`"w"`	Write (NEW)	Creating new file
`"a"`	Append (ADD)	Adding to existing file

Warning: `"w"` deletes everything in the file!

💡 Common Patterns

1. Count Rows

```
data = list(csv.DictReader(file))
row_count = len(data)
```

2. Sum Column

```
total = 0
for row in data:
    total += float(row['price'])  # Convert to number!
```

3. Find Max

```
max_value = 0
max_row = None

for row in data:
    value = float(row['revenue'])
    if value > max_value:
        max_value = value
        max_row = row
```

4. Search

```
def find_by_name(data, name):
    for row in data:
        if row['name'].lower() == name.lower():
            return row
    return None
```

❌ Common Mistakes

Reading:

```
# WRONG: Trying to use reader after file closes
with open("file.csv", "r") as file:
    reader = csv.DictReader(file)
# reader is now dead!

# RIGHT: Save to list
with open("file.csv", "r") as file:
    data = list(csv.DictReader(file))
# data lives on!
```

Writing:

```
# WRONG: Forgetting newline=""
with open("file.csv", "w") as file:  # ← Bad on Windows

# RIGHT: Always include it
with open("file.csv", "w", newline="") as file:  # ← Good
```

🎯 Quick Cheat Sheet

Reading:

```
import csv
with open("file.csv", "r") as file:
    data = list(csv.DictReader(file))
# Use: data[0]['column'], len(data)
```

Writing (New):

```
import csv
with open("file.csv", "w", newline="") as file:
    writer = csv.DictWriter(file, fieldnames=['a','b','c'])
    writer.writeheader()
    writer.writerows(data)
```

Writing (Add):

```
import csv
with open("file.csv", "a", newline="") as file:
    writer = csv.DictWriter(file, fieldnames=['a','b','c'])
    writer.writerow(new_row)  # NO header!
```

✅ Remember These!

1. **Import csv** - Always start with this
2. **Numbers are strings** - Use `int()` or `float()`
3. **"w" deletes** - Be careful with write mode
4. **newline=""** - Always include when writing
5. **Save to list** - If you need data later